

N.S. LABORATORY SDN. BHD.

SOLITE, 1000mg/g, Powder for use in drinking water

Active Ingredient:

Each gram contains

Sodium salicylate -----1000mg

Product Description:

White or almost white crystalline powder which can be dissolved in animal drinking water.

Dosage form:

Powder for use in drinking water

Pharmacodynamics:

Sodium salicylate is a non-steroid anti-inflammatory drug (NSAID) and has an anti-inflammatory, analgesic and antipyretic effect. The mode of action is based on inhibition of the enzyme cyclooxygenase, resulting in decreased production of prostaglandin (inflammation mediators).

Pharmacokinetics:

Orally administered sodium salicylate is rapidly absorbed by passive diffusion, partially from the stomach, but mainly from the anterior part of the small intestine. Sodium salicylate distributes very well to the various tissues. Values of volume of distribution (Vd) are higher in the newborns. Half lives are longer in the very young resulting in slower elimination of the substance. This is most prominent in animals up to 7-14 days of age. Metabolism takes place mainly in the endoplasmic reticulum and the mitochondria of the liver cells. Elimination occurs mainly via the urine and the pH of the urine can have a major effect on this elimination.

Indications:

Calves- supportive treatment of pyrexia in acute respiratory disease, in combination with appropriate (e.g. anti-infective) therapy if necessary.

Pigs- For the treatment of inflammation in combination with concurrent antibiotic therapy.

Mode of Administration:

Oral administration via drinking water.

Recommended Dosage:

Calves- 40 mg sodium salicylate per kg bodyweight once daily, for 1 to 3 days.

Administration: orally in drinking water.

Pigs- 35 mg sodium salicylate per kg bodyweight per day, for 3 to 5 days.

Administration: orally in drinking water.

The following formula can be used to calculate the concentration of product in drinking water:

$$\frac{x \text{ mg product/kg bodyweight/day} \times \text{mean bodyweight(kg) of animals to be treated}}{\text{Mean daily water consumption (l) per animal}}$$

= x mg product per l drinking water

Alternatively the product can also be administered with the drinking water as pulse medication. Half of the calculated total daily amount of powder is mixed with 5-10 litres of clean water and stirred until evenly dispersed. This solution is then added, whilst stirring, into an amount of drinking water that will be consumed within approximately 3-4 hours and administered twice daily.

Maximum solubility of the product in water is approximately 100 g/litre.

The use of suitably calibrated weighing equipment for the administration of the calculated amount of sodium salicylate is recommended.

Contraindications:

Do not administer in case of severe hypoproteinaemia, liver and kidney disorder.

Do not administer in case of gastrointestinal ulcerations and chronic gastrointestinal disorders.

Do not administer in case of malfunction of the haemopoietic system, coagulopathy, haemorrhagic diathesis.

Do not use sodium salicylates in neonates or calves less than 2 weeks of age.

Do not use in piglets of less than 4 weeks of age

Do not use in animals with known hypersensitivity to sodium salicylate.

Special Warnings for each Target Species:

None known.

Special Precautions for Use in Animals:

Given that sodium salicylate may inhibit clotting of blood, it is recommended that elective surgery should not be performed on animals within 7 days after the end of treatment.

Special Precautions to be taken by the Person Administering the Veterinary Medicinal Product to Animals:

People with known hypersensitivity to sodium salicylate or related substances (e.g. aspirin) should avoid contact with the veterinary medicinal product.

Irritation of the skin, eye, and respiratory tract might occur. During preparation and mixing of the product, direct contact with the skin and eyes and inhalation of the powder should be avoided. It is recommended to wear gloves, safety glasses, and a dust mask. In case of accidental dermal exposure wash skin immediately with water.

In the event of accidental eye contact, the user is advised to wash the eye with plenty of water for 15 minutes, and seek medical advice if irritation persists.

During administration of medicated water or milk (replacer) to the animals skin contact should be prevented by wearing gloves. Wash accidentally exposed skin immediately with water.

Interactions with other Medicaments:

Concurrent administration of potentially nephrotoxic drugs (e.g. aminoglycosides) should be avoided. Salicylic acid is highly plasma (albumin) bound and competes with a variety of compounds (e.g. ketoprofen) for plasma protein binding sites. Plasma clearance of salicylic acid has been reported to increase in combination with corticosteroids possibly due to induction of metabolism of salicylic acid. Concurrent use with other NSAIDs is not recommended, because of increased risk of gastro-intestinal ulceration. Drugs which affect blood clotting should not be used in combination with sodium salicylate.

Usage during Pregnancy and Lactation:

The use is not recommended during pregnancy and lactation because laboratory studies in rats have shown evidence of teratogenic and foetotoxic effects. Salicylic acid crosses the placenta and is excreted with the milk. Half-life in the newborn is longer and thus toxicity symptoms may occur much sooner. Furthermore platelet aggregation is inhibited and bleeding time increased which is not favourable during difficult parturition / caesarean section. Finally some studies indicate that parturition is postponed.

Adverse Effects:

Gastrointestinal irritation may occur especially in animals with pre-existing gastrointestinal disease. Such irritation may be clinically manifested by production of black faeces due to bleeding in the gastrointestinal tract. Inhibition of normal blood clotting may occur incidentally. This effect is reversible and diminishes within approximately 7 days.

Overdose and Treatment:

Calves tolerate dosages up to 80 mg/kg for 5 days or 40 mg/kg for 10 days without any adverse effects. Pigs tolerate dosages up to 175 mg/kg for up to 10 days without any significant adverse effects. In case of an acute overdose intravenous bicarbonate infusion results in a higher clearance of salicylic acid by alkalisation of the urine and may be beneficial in correcting (secondary metabolic) acidosis.

Withdrawal Periods (Meat and Offal):

Pigs- 0 days;

Calves- 0 days;

Do not use in cows producing milk for human consumption.

Storage Condition:

Store in cool, dark place (< 30°C). Protect from moisture.

Shelf Life:

24 months

Shelf Life after Opening:

Discard any unused powder immediately after opening

Shelf Life after Dilution/Reconstitution:

24 hours

Packaging Available:

100g/packet, 100packets/carton, 10kg/carton;

500g/packet, 30packets/carton, 15kg/carton

Manufacturer and Registration Holder:

N.S. Laboratory Sdn. Bhd.

54, Jalan S2C2, Green Technology Park, Seremban 2, 70300 Seremban, Negeri Sembilan, Malaysia.

Revised Date:

27/04/2022